STAT3888: Full Population Interaction Model

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This document will try to implement one idea I had a bit ago, which was to use the full population data set and introduce interaction effects between `EXREGUIC` and the alcohol and smoking filtering variables, instead of directly filtering on them. In particular, we define the variable

$$H = \begin{cases} 1, & \text{if smoker and medium/high risk alcohol consumer} \\ 0, & \text{otherwise} \end{cases}$$

and fit it and the interaction with `EXREGUIC` alongside it as well, on our usual multinomial logistic regression model, to see if association between physical activity and ALT/GGT is different in the high-risk population.

Adding H to the Dataset

Recall that we filtered based on `AL2K7DAY %in% c("Medium risk", "High risk")` and `SMKAGEDB != "Never smoked daily"`; we also filtered on age being >18 but that is already done in our `fdat.full`.

▼ Code

```
# Load dataset and models
fdat.full = readRDS(here("Objects", "fdat_full.rds"))
alt.fmlr = readRDS(here("Objects", "Full_data_MLR_ALT3.rds"))
ggt.fmlr = readRDS(here("Objects", "Full_data_MLR_GGT3.rds"))

# Then introducing H Filter
fdat.full.h = fdat.full %>%
  mutate(H = case_when(
    is.na(AL2K7DAY) | is.na(SMKAGEDB) ~ NA,
    AL2K7DAY %in% c("Medium risk", "High risk") &
      SMKAGEDB != "Never smoked daily" ~ 1,
    .default = 0))

# Proportions of H
table(fdat.full.h$H)
```

```
    0     1
```

```
13897  1362
```

The proportion of $H = 1$ coincides with the number of rows in our restricted `pdat.full`, so it seems that the implementation was alright.

Model Fitting

We fit the new models first, check convergence, AIC, and deviance, the overall fits and significance for the main variables `EXREGUIC` and `H`.

▼ Code

```
# ALT Interaction fit
alt.int = multinom(ALT3 ~ EXREGUIC * H + AGEB + SEX + WAIST + DS
    data = fdat.full.h, Hess = TRUE)
```

```
# weights:  75 (48 variable)
initial  value 5866.589621
iter  10 value 3733.601751
iter  20 value 3475.625933
iter  30 value 3445.295417
iter  40 value 3437.815821
iter  50 value 3432.076836
iter  60 value 3431.116852
final  value 3431.116792
converged
```

▼ Code

```
# GGT Interaction fit
ggt.int = multinom(GGT3 ~ EXREGUIC * H + AGEB + SEX + WAIST + DS
    data = fdat.full.h, Hess = TRUE)
```

```
# weights:  75 (48 variable)
initial  value 5866.589621
iter  10 value 3279.732827
iter  20 value 3035.545223
iter  30 value 2984.544775
iter  40 value 2981.579442
iter  50 value 2980.913598
iter  60 value 2980.311293
iter  70 value 2980.132392
final  value 2980.132159
converged
```

▼ Code

```
# Convergence, AIC, Deviance
conv = c(alt.int$convergence, ggt.int$convergence)
conv.log = (conv == 0)
models = c("ALT Interaction", "GGT Interaction")
aic = c(alt.int$AIC, ggt.int$AIC)
deviance = c(alt.int$deviance, ggt.int$deviance)
models.df = data.frame("Model" = models, "Convergence" = conv.l
                       "AIC" = aic, "Deviance" = deviance)
kable(models.df, align = "cccc")
```

Model	Convergence	AIC	Deviance
ALT Interaction	TRUE	6958.234	6862.234
GGT Interaction	TRUE	6056.264	5960.264

▼ Code

```
# Model summaries (for now, just EXREGUIC and H)
tidy(alt.int)[c(2, 3, 24, 26, 27, 48), ] %>%
  kable(caption = "ALT Interaction Effects", digits = 4, align =
        col.names = c("ALT Level", "Term", "Coefficient", "SE",
```

ALT Interaction Effects

ALT Level	Term	Coefficient	SE	t-score	p-value
Borderline	EXREGUICDid not meet recommended guidelines	0.0048	0.0970	0.0496	0.9604
Borderline	H	0.3516	0.2057	1.7092	0.0874
Borderline	EXREGUICDid not meet recommended guidelines:H	-0.6843	0.3340	-2.0489	0.0405
High	EXREGUICDid not meet recommended guidelines	0.2825	0.0957	2.9527	0.0032
High	H	0.7085	0.1952	3.6301	0.0003
High	EXREGUICDid not meet recommended guidelines:H	-0.5700	0.2847	-2.0019	0.0453

▼ Code

```
tidy(ggt.int)[c(2, 3, 24, 26, 27, 48), ] %>%
```

```
  kable(caption = "GGT Interaction Effects", digits = 4, align =
        col.names = c("GGT Level", "Term", "Coefficient", "SE",
```

GGT Interaction Effects

GGT Level	Term	Coefficient	SE	t-score	p-value
Borderline	EXREGUICDid not meet recommended guidelines	0.1931	0.1336	1.4453	0.1484
Borderline	H	0.8658	0.2457	3.5239	0.0004
Borderline	EXREGUICDid not meet recommended guidelines:H	-0.4004	0.3642	-1.0996	0.2715
High	EXREGUICDid not meet recommended guidelines	0.4378	0.0937	4.6698	0.0000
High	H	1.0900	0.1754	6.2129	0.0000
High	EXREGUICDid not meet recommended guidelines:H	-0.3266	0.2451	-1.3326	0.1827

Model Comparison

We just conduct some simple model comparison tests and metrics, like likelihood ratio, AIC, and BIC, and the pseudo-R^2 via `pscl::pR2`.

▼ Code

```
# Likelihood Ratio Test (we need to refit the MLR models using
# These should be identical though; but for some reason I am ge
# so I will further make the data set only complete cases
fdat.full.hc = fdat.full.h %>% drop_na()
alt.fmlr2 = multinom(formula = ALT3 ~ EXREGUIC + AGEB + SEX + W
    data = fdat.full.hc, Hess = TRUE, na.action = "na.omit")
```

```
# weights:  69 (44 variable)
initial  value 5866.589621
iter  10 value 3737.337822
iter  20 value 3480.767272
iter  30 value 3452.903491
iter  40 value 3444.696965
iter  50 value 3438.956055
final  value 3438.910955
converged
```

▼ Code

```
lrtest(alt.fmlr2, alt.int)
```

```
Likelihood ratio test

Model 1: ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB
Model 2: ALT3 ~ EXREGUIC * H + AGEB + SEX + WAIST + DSMKHHRB
  #Df  LogLik Df  Chisq Pr(>Chisq)
1  44 -3438.9
2  48 -3431.1  4 15.588   0.003624 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

▼ Code

```
ggt.fmlr2 = multinom(formula = GGT3 ~ EXREGUIC + AGEB + SEX + WA
    data = fdat.full.hc, Hess = TRUE, na.action = "na.omit")
```

```
# weights:  69 (44 variable)
initial  value 5866.589621
iter  10 value 3278.539328
iter  20 value 3055.881584
iter  30 value 3014.378981
iter  40 value 3011.226968
iter  50 value 3010.448351
iter  60 value 3009.862587
final  value 3009.850756
converged
```

▼ Code

```
lrtest(ggt.fmlr2, ggt.int)
```

```
Likelihood ratio test

Model 1: GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB
Model 2: GGT3 ~ EXREGUIC * H + AGEB + SEX + WAIST + DSMKHHRB
  #Df  LogLik Df  Chisq Pr(>Chisq)
1  44 -3009.8
2  48 -2980.1  4 59.437  3.809e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

▼ Code

```
# AIC and BIC
data.frame(Model = c("ALT", "ALT With Interaction",
                     "GGT", "GGT With Interaction"),
           AIC = AIC(alt.fmlr2, alt.int, ggt.fmlr2, ggt.int)$AI(
           BIC = BIC(alt.fmlr2, alt.int, ggt.fmlr2, ggt.int)$BI(
  kable(caption = "AIC and BIC of Models", align = "ccc", digits
```

AIC and BIC of Models

Model	AIC	BIC
ALT	6965.822	7255.473
ALT With Interaction	6958.234	7274.217
GGT	6107.702	6397.353
GGT With Interaction	6056.264	6372.247

▼ Code

```
# Pseudo R^2 approximates proportion of variance explained
# like R2 but for categorical variables. Use McFadden (higher is
mcfadden = data.frame(
  Model_Name = c("ALT", "ALT with Interaction", "GGT", "GGT with
  McFadden = round(c(
    pR2(alt.fmlr2)["McFadden"],
    pR2(alt.int)["McFadden"],
    pR2(ggt.fmlr2)["McFadden"],
    pR2(ggt.int)["McFadden"]
  ), 4)
)
```

```
fitting null model for pseudo-r2
# weights:  6 (2 variable)
initial  value 5866.589621
final  value 3729.577411
converged
fitting null model for pseudo-r2
# weights:  6 (2 variable)
initial  value 5866.589621
final  value 3729.577411
converged
fitting null model for pseudo-r2
# weights:  6 (2 variable)
initial  value 5866.589621
final  value 3173.547535
converged
```

```
fitting null model for pseudo-r2
# weights:  6 (2 variable)
initial  value 5866.589621
final  value 3173.547535
converged
```

▼ Code

```
row.names(mcfadden) = NULL

# 2. View the clean table in your console
kable(mcfadden, caption = "McFadden R^2 for Models (Higher is be
      align = "cc", col.names = c("Model", "McFadden"))
```

McFadden R^2 for Models (Higher is better)

Model	McFadden
ALT	0.0779
ALT with Interaction	0.0800
GGT	0.0516
GGT with Interaction	0.0609

We see overall based on the traditional model comparison metrics, the interaction models are overall “better”, at least compared to the full model. All four models explain a modest but definitive proportion of the outcome variability, with the interaction models offering a better statistical fit, particularly for `GGT`, which saw the largest relative improvement.